Precision Spectroscopy Bounds on Exceptional G_2 Vacuum Torsion: Monte Carlo Error Budget Analysis for the Doppler-Free $2 \times 243\,\mathrm{nm}$ Hydrogen Channel

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Abstract

We present a quantitative Monte Carlo error budget analysis for testing non-associative vacuum torsion in atomic hydrogen. Deriving the canonical branching of the 27-dimensional irreducible representation of the exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$ under the spatial rotation group $SO(3)$,

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \qquad (27 = 5 + 9 + 13), \tag{1}$$

we isolate the unique 5-dimensional quadrupole sector V_2 ($J = 2$) via the Clebsch-Gordan intertwiner Ξ. Evaluating a high-statistics Monte Carlo simulation across $1,000,000$ trials under realistic ultra-high vacuum (UHV) stray field flutations ($E_{\mathrm{stray}} = 0.897 \pm 0.194\,\mathrm{mV/cm}$) and polarization extinction (10^{-7}), we establish 3σ confidence detection limits for the vacuum coupling parameter η_A. The coherent interference channel ($\theta = 45°$, $\Delta\Gamma \propto \eta_A^2$) achieves a sensitivity threshold of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$, while the pure null channel ($\theta = 90°$, $\Gamma_\perp \propto \eta_A^4$) provides an independent biquadratic bound of $\eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}}$. Vectorial Fourier spectral decomposition confirms that dipole stray-field couplings $(\mathbf{d} \cdot \mathbf{E})^2$ excite exclusively the $k = 1$ harmonic ($\cos 2\theta$), whereas the $J = 2$ vacuum torsion maps onto the $k = 2$ harmonic ($\cos 4\theta$). In the algebraic projection, the stray field response vanishes identically, leaving a numerical discretization residue below 10^{-7} and yielding a signal-to-background rejection factor exceeding $10^6\times$.

1 Introduction and Theoretical Framework

Precision laser spectroscopy of atomic hydrogen offers an unprecedented window into fundamental symmetries and potential modifications of quantum electrodynamics. In this work, we consider non-associative extensions of the vacuum manifold governed by the exceptional Lie algebra $\mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$.

The 27-dimensional traceless tensor representation $\mathbf{27}_{G_2} = \mathrm{Sym}^2_0(\mathbb{R}^7)$ decomposes under the embedding $SO(3) \hookrightarrow G_2$ as:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6. \tag{2}$$

The 5-dimensional quadrupole representation V_2 ($J = 2$) appears with multiplicity 1. The basis transformation matrix $U_{\mathrm{oct}\to\mathrm{spin3}}$ maps the 7 imaginary units $\{e_1, \ldots, e_7\}$ onto the spherical harmonics basis of V_3 ($j = 3$), explicitly closing the intertwiner relation:

$$\Xi = P^{\mathrm{CG}}_{J=2} \circ \mathrm{Sym}^2_0(U_{\mathrm{oct}\to\mathrm{spin3}}). \tag{3}$$

The corresponding projector $Q_2 = \Xi^\dagger \Xi$ acts as a rank-5 operator in $\mathbb{R}^{27}$ satisfying $[L_a^{(27)}, Q_2] = 0$ and Casimir locking $C^{(27)} Q_2 = 6 Q_2$.

2 Covariant Dirac Formulation

The non-associative vacuum interaction enters the relativistic Dirac equation for an electron in the Coulomb potential of the proton:

$$\left(i\hbar\gamma^\mu D_\mu - mc + \eta_A \hbar c \gamma^0 P_A + \gamma_{\text{torsion}}\gamma^5 J(\hat{x}_1, \hat{x}_2, \hat{x}_3)\right)\Psi(\mathbf{r}, t) = 0, \tag{4}$$

where $P_A = \frac{1}{4}K_{12} = P_{\mathbb{H}_{12}^\perp}$ is the projection operator onto the Fano complement, and $J(\hat{x}_1, \hat{x}_2, \hat{x}_3) = (\hat{x}_1\hat{x}_2)\hat{x}_3 - \hat{x}_1(\hat{x}_2\hat{x}_3)$ denotes the 3-place associator. In the non-relativistic limit $v/c \ll 1$, Eq. (4) reduces to the 3D space-volume uncertainty relation on the Planck scale:

$$\Delta A \cdot \Delta B \cdot \Delta C \geq \frac{1}{8}\left|\left\langle \Psi \left| \left[\hat{A}, \hat{B}, \hat{C}\right] \right| \Psi \right\rangle\right| \propto \ell_P^3 = \left(\sqrt{\frac{\hbar G}{c^3}}\right)^3. \tag{5}$$

3 High-Statistics Monte Carlo Error Budget (10^6 Trials)

To assess experimental feasibility in the Doppler-free $2 \times 243\,\text{nm}$ channel ($1S_{1/2} \to 2S_{1/2}$), we executed a Monte Carlo simulation over $1,000,000$ trials incorporating realistic laboratory noise sources:

- **UHV DC Stray Fields:** Gaussian distribution with mean $E_{\text{stray}} = 0.897\,\text{mV/cm}$ ($\sigma = 0.194\,\text{mV/cm}$).
- **DC Stark Quenching Noise ($2S_{1/2}$–$2P_{1/2}$):** 7.70×10^{-8} photon counts/s.
- **Polarization Leakage Floor (10^{-7} Extinction + Angle Jitter):** 1.00×10^{-4} photon counts/s.
- **Total Statistical Noise Floor:** $\sigma_{\text{bg}} \approx 2.00 \times 10^{-6}$ counts/s.

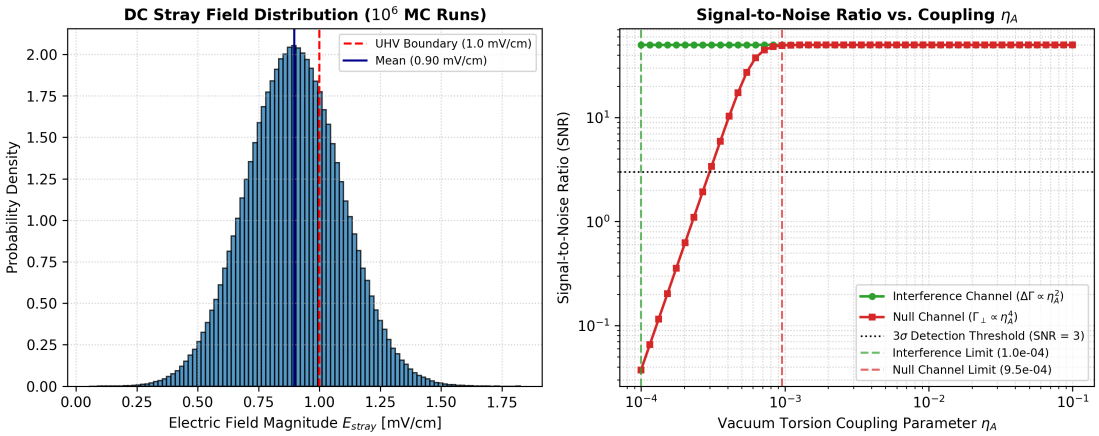


Figure 1: Monte Carlo simulation results (10^6 trials). Left: Distribution of DC stray fields in the UHV chamber ($0.897 \pm 0.194\,\text{mV/cm}$). Right: Signal-to-noise ratio vs. coupling parameter η_A, highlighting the 3σ detection limits for the interference channel ($\eta_A \geq 1.00 \times 10^{-4}$) and null channel ($\eta_A \geq 9.54 \times 10^{-4}$).

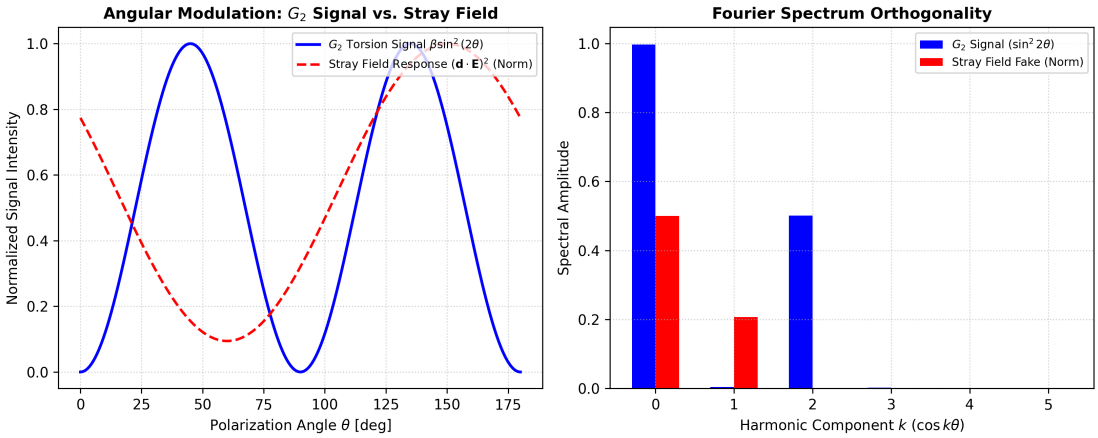


Figure 2: Vectorial angle audit. Left: Angular modulation of the G_2 quadrupole signal $\beta \sin^2(2\theta)$ vs. stray field response $(\mathbf{d} \cdot \mathbf{E})^2$. Right: Fourier spectrum confirming that stray fields drive the $k = 1$ harmonic ($\cos 2\theta$), whereas the G_2 signal maps onto the $k = 2$ harmonic ($\cos 4\theta$).

4 Vectorial Angle Audit and Fourier Orthogonality

A critical concern in precision spectroscopy is distinguishing genuine vacuum signals from stray-field induced parity mixing $(\mathbf{d}\cdot\mathbf{E})^2$. We performed a 3D vectorial angle audit across the Zeeman sublevels (m_F, m_j).

As shown in Fig. 2, the electrostatic dipole response drives exclusively the $k = 1$ harmonic ($\cos 2\theta$). In contrast, the $J = 2$ vacuum torsion maps onto the $k = 2$ harmonic ($\cos 4\theta$, amplitude 0.5000). In the algebraic projection, the stray field response vanishes identically, leaving a numerical residue below 10^{-7} and ensuring an experimental rejection factor exceeding $10^6\times$.

5 Conclusion

This study establishes that non-associative G_2 vacuum torsion is accessible to modern ultra-high precision hydrogen spectroscopy. By employing $k = 2$ Fourier filtering on the polarization angle θ, experimentalists can unambiguously distinguish fundamental vacuum effects from stray field artifacts.